

LINEAR REGRESSION FOR MULTIPLES

One cross-section, then twenty-five of them

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PRESENTATIONS

Three Groups

- 1 What your backtest earned, and what you compared it against
- 2 What changed when you switched to `ARY` and `datekey`
- 3 The thing that broke

If your annual return went down when you fixed the point-in-time problem, that is the most useful slide anyone will show today. Say by how much.

STOP BORROWING, START PREDICTING

What the Peer Median Assumes

That every company in an industry deserves the same multiple.

Two software companies, same industry. One grows at twenty percent with a forty percent margin; the other grows at three percent and barely breaks even. The peer median says they should trade at the same P/E, and calls the difference mispricing.

It is not mispricing. It is the model being wrong.

Borrow a multiple

Find companies that look like this one. Use the middle of what they trade at.

Predict a multiple

Estimate what the market pays for growth, for profitability, for leverage. Apply those prices to this company.

The second one is a regression, and the first one is a special case of it — a regression on industry dummies and nothing else.

The Alternative

ONE CROSS-SECTION

Freeze a Date

One row per company, on a single day. Three thousand rows, and no time dimension at all.

TELL GLM

Build a cross-section as of June 30 of last year: one row per stock, with P/E from **daily**, and from the most recent **sf1** annual filing bring in ROE, revenue growth, the operating margin, leverage, and payout. Add sector.

Everything today happens inside this one table. The time series comes back at the end.

What Goes on the Left

P/E

Bounded below at zero, unbounded above, and violently skewed. OLS on this is mostly fitting the tail.

log P/E

Roughly symmetric. Coefficients read as percentage effects on the multiple, which is how you want to think about them anyway.

And the negative-earnings problem does not go away — you cannot take the log of a negative P/E, so those companies leave the sample. Note how many, and which kind.

What Goes on the Right

Growth

Revenue growth, earnings growth — what the multiple is supposed to be paying for

Profitability

ROE, ROA, margins

Risk

Leverage, and how variable the earnings have been

Size

Log market cap

Sector

Dummies — which is where the peer median hides

Start with a handful you can defend. A regression with forty right-hand variables is a machine learning problem, and that is Tuesday.

Run It

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Regress log P/E on those characteristics with sector fixed effects. Winsorize the right-hand variables at 1 and 99 percent. Show me the coefficients, the standard errors, and the R-squared.

Then read the coefficients before you look at anything else. Does the market pay more for growth? For profitability? If a sign is backwards, that is the finding, and it usually means a variable is built wrong.

What R-Squared to Expect

Somewhere between a third and a half. Which is enormous, and does not mean what you want it to mean.

Cross-sectional valuation regressions fit well because companies genuinely do trade on their fundamentals. The question is not whether the fit is good. It is whether what is left over is worth trading.

RELATIVE VALUE

Fitted

What the market, on average, pays for a company with these characteristics.

Actual

What this company actually trades at.

Residual

The gap. Negative means it trades below what its own characteristics justify.

The residual is the signal, and it is exactly the same shape as last session's signal — own multiple against a predicted one. All that changed is how the prediction is made.

Fitted and Residual

Two Readings of a Negative Residual

IT IS CHEAP

The market has not noticed. Buy it, wait for the gap to close.

This is the assumption the whole strategy rests on.

YOUR MODEL IS MISSING SOMETHING

The company is in litigation, or its growth is about to stop, and the market knows. The gap is information you failed to include.

You cannot tell these apart for any single company. You can only find out in aggregate, over many companies and many years, which is what the backtest is for.

Look at the Extremes

TELL GLM

Show me the twenty largest positive and twenty largest negative residuals, with their characteristics and their sector.

If the cheap list is all one sector, your sector dummies are not doing their job. If it is all microcaps, size is entering the pricing in a way your model has not captured. Either way you learn it in thirty seconds by looking.

Against Last Session

The peer median is the same regression with only sector dummies on the right.

TELL GLM

Fit it three ways on this cross-section: sector dummies only, characteristics only, and both. Compare the R-squared and the correlation between the three sets of residuals.

If the residuals correlate at 0.9, the extra machinery is not buying you a new signal. It is buying you the same signal with more steps.

TWENTY-FIVE CROSS-SECTIONS

One regression per year, using only that year's data.

The temptation is to fit once on everything and apply the coefficients throughout. That uses 2024 data to price a 2004 company. It is the same look-ahead as **MRY**, wearing a different hat, and it is harder to spot because nothing about the code looks wrong.

Refit Every Year

The Loop

- 1 For each June 30 from 2001 to 2025, build the cross-section
- 2 Fit the regression on that year alone
- 3 Take the residuals
- 4 Sort into deciles, long the cheapest, short the most expensive
- 5 Hold twelve months, record the return

Step 2 is the only thing that changed from Tuesday. Everything else is the backtest you already have — which is the point of having written it as code you can read.

Watch the Coefficients Move

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Plot each coefficient by year, with its confidence band.

The price of growth is not a constant. If a coefficient flips sign in 2009 and again in 2021, that is worth a slide in your presentation — and it is a reason to be suspicious of any model that assumes the relationship is fixed.

How Many Companies Survive Each Year

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For each year, report how many stocks had all the variables and how many were dropped, broken out by which variable was missing.

Every characteristic you add shrinks the sample, and it does not shrink it randomly. A model that requires five years of earnings history is a model estimated on established companies only.

DID IT PAY

Peer median	Tuesday's result
Regression residual	Today's
Plain P/E	No prediction at all — just buy the low multiples

Report the average annual return, the volatility, the t-stat, and the worst year for all three. Same dates, same universe, same weighting, or the comparison means nothing.

Three Backtests, One Table

The Honest Outcome

It is entirely possible that plain P/E wins.

That would be a real result, and a well-known one. The elaborate model has more ways to be wrong, more parameters to fit to noise, and a smaller sample to fit them on. Report it if you find it.

What you must not do is keep trying specifications until one of them beats the others, and then present that one.

BEFORE TUESDAY

Three Things

- 1 Get the annual refit loop running, and produce the three-way comparison table
- 2 Save the cross-section builder as a skill — you will use it every week from here
- 3 Write the memory file, including which specifications you tried and rejected

Tuesday we replace the regression with something that can find relationships you did not specify. Come with the linear result written down, because it is the benchmark that machine learning has to beat.